# Cubean Coffee Roaster Modification

HY2.0 to USB-A Female Cable DIY Guide (V3 Final)

**⚠️ ABSOLUTE WARNING: DO NOT CONNECT CABLES ACCORDING TO THEIR COLORS!**

**The pre-installed HY2.0 male pigtail cable extending from the Cubean motherboard and any aftermarket HY2.0 cables have RANDOM color sequences!** If you intuitively connect red to red or black to black thinking they are 5V and GND, you will highly likely burn out the motherboard or your external device instantly.

**Completely ignore all wire colors. You MUST align and wire the connections strictly according to the physical order shown in "接線注意.jpg" (+5V, TX, RX, GND)!**

## 🛠 Materials Required


1. HY2.0 4-pin Female Pigtail Cable
(To connect with the male pigtail extending from the roaster)

2. USB-A Female Pigtail Cable

3. Tools: Wire strippers, soldering iron, solder, heat shrink tubing (or electrical tape).

## 📌 Pin Definition & Physical Mapping

There is no need to open the chassis or touch the onboard sockets directly. You only need to map your connections to the **HY2.0 male pigtail extending from the machine**. Please refer to the physical sequence below:



Internal Board Silkscreen:
**+5V, TX, RX, GND**

**Male Pigtail Extending from Roaster**
Ignore wire colors; use this physical order.

Standard USB-A internal wire color definitions (highly recommended to verify with a multimeter):
**Red = VCC (5V), Black = GND, White = D-, Green = D+**

| Physical Sequence of the Roaster's Male Pigtail | DIY Cable Side (USB-A Female End) | Wiring Notes |
|---|---|---|
| **+5V** | VCC (Typically Red) | Power line. Absolutely critical to get this right. |
| **GND** | GND (Typically Black) | Ground line. |
| **TX** | D- or D+ (Data Line) | Depending on the circuit design of your external controller (e.g., ESP32-S3 board), TX/RX may need to be mapped straight or crossed with D+/D-. It is recommended to perform a temporary touch-test to verify data transmission before final soldering. |
| **RX** | D+ or D- (Data Line) | |

## 🛠 Assembly & Wiring Steps

1. **Align the Pins:** Bring your HY2.0 female connector and the roaster's extending HY2.0 male connector together. **Completely ignore the wire colors of both sides.** Cross-reference with "接線注意.jpg" to trace which four wires on your female pigtail actually correspond to +5V, TX, RX, and GND.
2. **Wire Prep:** Strip approximately 5mm of insulation from the ends of both the HY2.0 female pigtail and the USB female pigtail wires, then tin the exposed copper tips with a small amount of solder.

3. **Insert Heat Shrink:** Slide pieces of heat shrink tubing onto one side of the wires before soldering.
4. **Solder the Connections:** Solder the corresponding wires together according to the pin mapping table above. Once cooled, slide the heat shrink tubing over the exposed solder joints and apply heat (using a heat gun or lighter) until they shrink securely to prevent short circuits.
5. **Plug & Test:** Plug your newly made adapter cable directly into the **HY2.0 male pigtail extending from the Cubean roaster**. Connect your external smart display or controller (such as an ESP32-S3 module) to the USB-A female side to verify that both power and data work correctly.